DESCRIPTION OF COURSEWORK

Course Code	EEE305
Course Name	Embedded Systems
Lecturer	Dr. Tan Jian Ding
Academic Session	2024/04
Assessment Title	Case Study (Group Work)

A. Introduction/ Situation/ Background Information

Embedded systems have a vast variety of application domains that varies from low cost to high, consumer electronics to industrial equipment, entertainment devices to academic equipment and medical instruments to weapons and aerospace control systems. In this assessment, students are required to analyze some of the embedded system malfunctions which leads to even bigger failures in aerospace applications.

B. Course Learning Outcomes (CLO) covered

This assignment is designed to cover the CLOs as followed:

CLO 1 Demonstrate knowledge on fundamental embedded systems.

CLO 3 Analyse the techniques of embedded systems architecture.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalized by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

1. This case study will contribute to 20% of your total assessment for this subject.
2. In this group work, you are required to develop an investigative report on a case study, and to conduct a 5-minute presentation on the studied case. The case that you will study can be of any case of your choice, but it has to fulfill ALL of the following criteria:
    a. It has to be a case of aerospace malfunction/disaster (inner space or outer space, your choice).
    b. The malfunction/disaster has to be due to a failure in the embedded systems.
    c. The selected case should be a specific case.
    d. The selected case has to be a true event that actually happened.

    Example: On July 4th, 1997, the Mars Pathfinder (an autonomous mobile robot) successfully set foot on the Martian surface. However, a few days after the start of its mission, the vehicle started losing information due to several system resets. The cause of this problem was located in the embedded systems kernel used in the Pathfinder.

    The resources and details of the case study will mainly be from the internet. Choose your case wisely.
3. Each group is required to submit one (1) report and conduct a 5-minute presentation on the case study in week 14.

4. The report and presentation should consist the following:
    a. A brief history on the incident/chronology of related events.
    b. Analysis on the cause of the failure.
    c. The suggestion from you, as an engineer, on how to rectify and improve this.
    d. Conclusion.
    e. References.
5. Report format:
    a. Font: Times New Roman
    b. Font Size: Non-specified
    c. Line Spacing: Non-specified (up to you)
    d. Minimum Pages: Non-specified
    e. Maximum Pages: 5 (contents)
    f. Margin: Non-specified
6. Presentation format:
    a. Presentation time: 5 minutes
    b. Question and answer, if any: 5 minutes
    c. Slides format: Power Point (PPT)
7. Submit the report by **June 30th 2024** 2359H. The presentation is tentatively set at **July 1st 2024**. It seems like a long period, but kindly do not wait for the last minutes. You may submit the report earlier.
8. No hard copy submission needed. Submit online via Moodle in the form of Word Document. Name your file as "*Case Study Report: Group #*" with # being your group number.

E. Report Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Incident description	20
2.	Failure analysis	30
3.	Improvement suggestion	30
4.	Conclusion and references	10
5.	Overall Writing Skills	10
	TOTAL	**100**

F. Presentation Evaluation Breakdown

No.	Component Title	Percentage (%)
6.	Slides preparation	20
7.	Information conveying	50
8.	Time management	10
9.	Overall presentation flow	20
	TOTAL	**100**

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